CS3511 Midterm

Name: _____ GTID: _____

1 Short answer

Explain why your answers are correct.

1. (8 pts) Does every algorithm that runs in expected-time $O(T(n))$ also run in worst-case time $O(T(n))$?

2. (8 pts) Consider the recurrence

$$T(n) \leq T(n-1) + n \qquad \text{for } n \geq 2,$$

with initial condition

$$T(1) = c.$$

What is $O(T(n))$? What changes if $T(n) \leq T(n-3) + 2n$?

3. (12 pts) Dijkstra's algorithm works on directed graphs with non-negative edge-weights. Give an example graph with negative edge weights on which Dijkstra's algorithm fails.

4. (15 pts) Compute the Fourier transform of this polynomial: $1 + 2t - t^2 + t^3$. Do this by writing the Fourier transform as a $4x4$ matrix, and the calculating the matrix-vector product. Show your work.

5. (10 pts) There are many uses for the FFT. Describe two.

2 Algorithm design

1. Let $A[1 \dots n]$ be an array of n integers (there may be positive and negative values). A contiguous subarray of A is defined to be $A[i, i+1, \dots, j], i \leq j$. We are interested in algorithms that calculate values for i and j that minimize this sum.

   (a) (32 pts) Give a divide and conquer algorithm that finds the subarray of A with minimum sum that runs in time $\mathcal{O}(n \log n)$. Explain why it works and why it has this run-time.

(b) (33 pts) There is a dynamic programming approach that solves this problem in linear time. Design a linear-time algorithm for this problem. Explain why it works and why it runs in linear time. Hint: Consider subproblems, $b[i]$, where $b[i]$ equals the smallest-sum subarray ending at $A[i]$.

2. A company manages a network of potentially faulty computers. Each computer can send messages to some of the other computers, but not necessarily receive information back. That is, the communication channels may be directed.

   (a) (8 pts) Give an example of 6 computers, with communication channels, that requires at least 2 additional channels to be added.

(b) (20 pts) Give an efficient algorithm to determine the minimum number of additional connections necessary so that every computer can send information to every other computer. Show that the algorithm is correct. Explain its run-time.

Additional work space